STA 711 Final Exam Review

Information: The final exam will be **cumulative**, and will cover material from homeworks 1 - 10 and class up through Holm's method for controlling FWER. (FDR, Benjamini-Hochberg, and interim analysis topics will not appear in the main questions, though they could potentially appear in extra credit).

While you are expected to know material from the full course, the exam will place somewhat more emphasis on material since the second exam (hypothesis testing and confidence intervals).

The practice questions below mostly focus on this new material. For practice with earlier material, see the homework assignments and review questions from the first two exams. As always, the review questions are not completely comprehensive, but will give you a sense of the kinds of questions I could ask on the exam.

Resources and rules:

- You will be provided with a sheet of common distributions, including their probability function, mean, and variance. You may use any of the facts on this sheet without proof, unless specifically asked to derive them
- You will also be provided with a partial list of results from class and homework assignments (see the next page); you may use these results without proof, unless asked to specifically to derive one of these results or otherwise stated.
- Finally, any other results from class or homework may also be used without proof, unless asked to specifically to derive one of these results or otherwise stated.

Provided results

Quantile estimation

Let $Y_1, ..., Y_n \overset{iid}{\sim} F$, for some continuous F with pdf f, and let θ_0 be the pth quantile of F, and $\widehat{\theta}$ the sample pth quantile. Then,

$$\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \frac{p(1-p)}{f^2(\theta_0)}\right)$$

(provided that $f(\theta_0) > 0$).

M-estimation (including extension)

Let $Y_1, ..., Y_n$ be iid samples from some distribution g. (Y_i could be a scalar or vector random variable; and in the regression setting, we typically have $Y_i = (\mathbf{x}_i, Y_i)$). Let $\psi(Y, \boldsymbol{\theta})$ be a function which depends on a parameter $\boldsymbol{\theta} \in \mathbb{R}^d$.

Let $\widehat{\boldsymbol{\theta}}$ solve

$$\sum_{i=1}^{n} \psi(Y_i, \boldsymbol{\theta}) = \boldsymbol{c}_n$$

where $\boldsymbol{c}_n/\sqrt{n} \xrightarrow{p} \mathbf{0}$ as $n \to \infty$ (in the simplest case, we just have $\boldsymbol{c}_n \equiv \mathbf{0}$). And let $\boldsymbol{\theta}_0$ solve

$$\mathbb{E}_g[\psi(Y_i, \boldsymbol{\theta})] = \mathbf{0}$$

Then, under regularity conditions (which you may assume hold unless otherwise specified), $\widehat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$, and

$$\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S}(\boldsymbol{\theta}_0)) \qquad \mathbf{S}(\boldsymbol{\theta}_0) = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$$

where $\mathbf{B} = \mathbb{E}_g[\psi(Y_i, \boldsymbol{\theta}_0)\psi(Y_i, \boldsymbol{\theta}_0)^T]$. The definition of $\mathbf{A}$ depends on whether ψ is differentiable. If ψ is differentiable, then

$$\mathbf{A} = -\mathbb{E}_g\left[\psi'(Y_i, \boldsymbol{\theta}_0)\right]$$

with $\psi'(Y_i, \boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}}\psi(Y_i, \boldsymbol{\theta})$. Otherwise,

$$\mathbf{A} = -\frac{\partial}{\partial \boldsymbol{\theta}}\mathbb{E}_g[\psi(Y_i, \boldsymbol{\theta})]\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}$$

PSD matrices

Suppose that $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is a multivariate normal random vector. We know that the covariance matrix $\boldsymbol{\Sigma}$ is positive semi-definite, and a useful fact about positive semi-definite matrices is that there exists a unique positive semi-definite matrix, which we will denote $\boldsymbol{\Sigma}^{\frac{1}{2}}$, such that $\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{\Sigma}^{\frac{1}{2}} = \boldsymbol{\Sigma}$. It is also true that $\boldsymbol{\Sigma}^{-\frac{1}{2}} := (\boldsymbol{\Sigma}^{\frac{1}{2}})^{-1}$ satisfies $\boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{\Sigma}^{-\frac{1}{2}} = \boldsymbol{\Sigma}^{-1}$.

t distribution

Let $Z \sim N(0, 1)$ and $V \sim \chi^2_d$ be *independent*. Then,

$$\frac{Z}{\sqrt{V/d}} \sim t_d$$

Sample variance and χ^2 distribution

Let $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, and let

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$$

be the sample variance. Then,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

Cochran's theorem

Theorem: Let $Z_1, ..., Z_n \overset{iid}{\sim} N(0,1)$, and let $\mathbf{z} = (Z_1, ..., Z_n)^T$. Let $\mathbf{A}_1, ..., \mathbf{A}_k \in \mathbb{R}^{n \times n}$ be symmetric matrices such that $\mathbf{z}^T\mathbf{z} = \sum_{i=1}^{k} \mathbf{z}^T\mathbf{A}_i\mathbf{z}$, and let $r_i = rank(\mathbf{A}_i)$. Then the following are equivalent:

- $r_1 + \cdots + r_k = n$
- The $\mathbf{z}^T\mathbf{A}_i\mathbf{z}$ are independent
- Each $\mathbf{z}^T\mathbf{A}_i\mathbf{z} \sim \chi^2_{r_i}$

Questions

1. Let $X_1, ..., X_n$ be an iid sample from a distribution with pdf

$$f(x|\theta) = \frac{2}{\sqrt{\theta\pi}} \exp\left\{-\frac{x^2}{\theta}\right\} \qquad x > 0,\ \theta > 0.$$

You may use, without proof, the fact that $\dfrac{2X_i^2}{\theta} \sim \chi_1^2$.

(a) Show that the likelihood ratio test of $H_0 : \theta = \theta_0$ vs. $H_A : \theta \neq \theta_0$ rejects when $\dfrac{2\sum_{i=1}^n X_i^2}{\theta_0} < c_1$ or $\dfrac{2\sum_{i=1}^n X_i^2}{\theta_0} > c_2$, for some $c_1 < c_2$.

(b) Find values c_1, c_2 such that the likelihood ratio test from (a) is a size α test.

2. Let $Y_1, ..., Y_n \overset{iid}{\sim} Poisson(\lambda)$, and let $\tau = P(Y_i = 0)$. We have previously shown that the MLE for λ is $\widehat{\lambda}_{\text{MLE}} = \overline{Y}$.

(a) Find the MLE $\widehat{\tau}_{\text{MLE}}$ for τ.

(b) Now consider another estimator: $\widetilde{\tau} = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{Y_i = 0\}$. Find the asymptotic relative efficiency $ARE(\widehat{\tau}_{\text{MLE}}, \widetilde{\tau})$.

(c) We are interested in testing $H_0 : \tau = \tau_0$ vs. $H_A : \tau \neq \tau_0$. Propose two different Wald tests for these hypotheses (specify when you will reject), one based on $\widehat{\tau}_{\text{MLE}}$ and one based on $\widetilde{\tau}$. Which test will be more powerful?

3. **(Multivariate normal means)** Let $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$ be two independent multivariate normal random vectors, with

$$\mathbf{x}_1 \sim N(\boldsymbol{\theta}_1, \sigma^2\mathbf{I}_d) \qquad \mathbf{x}_2 \sim N(\boldsymbol{\theta}_2, \sigma^2\mathbf{I}_d),$$

where $\mathbf{I}_d$ is the $d \times d$ identity matrix.

(a) Show that $\mathbf{x}_2 - \mathbf{x}_1 \sim N(\boldsymbol{\theta}_2 - \boldsymbol{\theta}_1, 2\sigma^2\mathbf{I}_d)$.

(b) Assume that σ^2 is *known*, but $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$ are unknown. We wish to test $H_0 : \boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$ vs. $H_A : \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$. Using part (a), propose a test statistic that follows a χ_d^2 distribution under H_0, and describe when you will reject for a level α test. (Note: the χ^2 distribution here should be *exact*, not asymptotic; this is possible because the underlying data are normal).

(c) Still assuming that σ^2 is known, under H_A the test statistic from part (b) should follow a non-central chi-squared distribution $\chi_d^2(\lambda)$. Give the non-centrality parameter λ, and write down the power function.

(d) Finally, suppose now that σ^2 is *unknown*, but it is known that

$$\boldsymbol{\theta}_2 = \boldsymbol{\theta}_2 + \begin{pmatrix} \delta \\ \delta \\ \vdots \\ \delta \end{pmatrix}$$

for some $\delta \in \mathbb{R}$. (That is, $\theta_{2,j} = \theta_{1,j} + \delta$ for each entry $j = 1, ..., d$). For the test of $H_0 : \delta = 0$ vs. $H_A : \delta \neq 0$, propose a test statistic which has a t_{d-1} distribution under H_0.

4. **(Estimation in the geometric model)** For this problem, a geometric random variable $X \sim Geometric(\theta)$ with $\theta \in (0,1)$ has pmf

$$f(x|\theta) = P_\theta(X = x) = (1-\theta)^x \theta \qquad x \in \{0, 1, 2, ...\}$$

It will also help to know that

$$\mathbb{E}_\theta[X] = \frac{1-\theta}{\theta} \qquad Var_\theta(X) = \frac{1-\theta}{\theta^2}$$

(This may be a slightly different parameterization than appears on the formula sheet).

Suppose that $X_1, ..., X_n \overset{iid}{\sim} Geometric(\theta)$.

(a) Find the maximum likelihood estimator $\widehat{\theta}$.

(b) Show that $\widehat{\theta} \xrightarrow{p} \theta$. (While it is true that MLEs are generally consistent, I do not want you to use that general fact here; rather, I want you to show it directly using your estimator from part (a)).

(c) Find the asymptotic distribution of $\sqrt{n}(\widehat{\theta} - \theta)$. You may use our usual asymptotic results, and assume that any necessary regularity conditions hold.

(d) Use part (c) to construct a Wald confidence interval for θ.

5. **(Six Gaussians)** Let $X_1, ..., X_6$ be independent univariate normal random variables with $X_i \sim N(\theta_i, \sigma^2)$.

(a) For this part of the problem **only**, assume that $\sigma^2 = 1$ is known, and we wish to test the null hypothesis $H_0 : \theta_2 = \theta_3$ and $\theta_4 = \theta_5 = \theta_6$, vs. the alternative that $\boldsymbol{\theta} = (\theta_1, ..., \theta_6)^T$ is any other vector in $\mathbb{R}^6$. Suggest a test statistic which follows a χ^2 distribution, and specify the degrees of freedom.

Hints:

- You should not use any large-sample approximations here. Because the X_i are normal, it is possible to get a test statistic which is *exactly* χ^2.
- Think carefully about the different tests we have discussed, and choose one that allows you to specify these restrictions under H_0.

(b) Now, for this part of the problem **only**, instead assume that σ^2 is unknown, but it *is* known that $\theta_2 = \theta_3$ and $\theta_4 = \theta_5 = \theta_6$. We are interested in the quantity $\lambda = \theta_4 - \theta_3$.

i. Write down an estimator for λ in terms of $X_1, ..., X_6$.

ii. Write down an estimator for σ^2 in terms of $X_1, ..., X_6$.

iii. Suggest a confidence interval based on the t distribution for the quantity $\lambda = \theta_4 - \theta_3$. Make sure to specify the appropriate degrees of freedom.

6. **(McNemar's Test)** Suppose we have paired binary data: for $i = 1, ..., n$ we observe (X_i, Y_i) with $X_i, Y_i \in \{0, 1\}$. The pairs (X_i, Y_i) are iid, with probabilities

$$P[(X_i, Y_i) = (a, b)] = \pi_{a,b} \qquad a, b \in \{0, 1\}$$

For instance, to borrow an example from Agresti's *Categorical Data Analysis*, consider a survey of $n = 1600$ British voters, each of whom was asked their opinion of the British prime minister (approval or disapproval), and then asked again six months later. In this example,

X_i is the ith voter's initial response ($X_i = 0$ indicates disapproval, $X_i = 1$ indicates approval), while Y_i is the same voter's subsequent response to the six-month follow-up question.

Let $\pi_{1+} = P(X_i = 1) = \pi_{10} + \pi_{11}$ and $\pi_{+1} = P(Y_i = 1) = \pi_{01} + \pi_{11}$. We are interested in testing the null hypothesis that $\pi_{1+} = \pi_{+1}$. Since X_i and Y_i are correlated (e.g. they are two responses from the same British voter), we can't use methods for two independent samples.

This is equivalent to testing

$$H_0 : \pi_{10} = \pi_{01} \qquad H_A : \pi_{10} \neq \pi_{01}$$

A common test of these hypotheses is *McNemar's test*. Letting $n_{01} = \sum_{i=1}^{n} \mathbb{1}\{X_i = 0, Y_i = 1\}$ and $n_{10} = \sum_{i=1}^{n} \mathbb{1}\{X_i = 1, Y_i = 0\}$, McNemar's test is based on the statistic

$$Z = \frac{n_{10} - n_{01}}{(n_{10} + n_{01})^{1/2}}$$

Under H_0, $Z \approx N(0, 1)$.

Really what this comes down to is testing for symmetric off-diagonals of a 2×2 contingency table. For example, for the British opinion poll example, the survey results are

First Survey \Second Survey	Disapprove	Approve	Total
Disapprove	570	86	656
Approve	150	794	944
Total	720	880	1600

and the test statistic is

$$Z = \frac{150 - 86}{(150 + 86)^{1/2}} = 4.17$$

The purpose of this question is to derive McNemar's test.

(a) Let $p_{10} = n_{10}/n$ and $p_{01} = n_{01}/n$ be the sample proportions, and let $D = p_{10} - p_{01}$. Using the fact that $Cov(n_{10}, n_{01}) = -n\pi_{10}\pi_{01}$ (you do not need to prove this covariance), find an expression for $Var(D)$.

(b) Suppose that the null hypothesis $H_0 : \pi_{10} = \pi_{01}$ is true, and let $\pi_1 = \pi_{10} = \pi_{01}$ be the common parameter under the null. Show that, under H_0,

$$\sqrt{n}D \xrightarrow{d} N(0, 2\pi_1)$$

(c) Show that, under H_0, $\dfrac{n_{10} - n_{01}}{(n_{10} + n_{01})^{1/2}} \approx N(0, 1)$.

7. Suppose that $X_{ij} \sim Poisson(\lambda_{ij})$ for $i, j \in \{0, 1\}$ are independent random variables. Further suppose that $\lambda_{ij} = \lambda_0 \rho^{i+j}$ where $\lambda_0, \rho > 0$ and λ_0 is known. Suggest a uniformly most powerful test of $H_0 : \rho \leq \rho_0$ vs. $H_A : \rho > \rho_0$. You do not need to give an explicit cutoff for the test, but you should give an explicit formula for the test statistic and explain why your test is UMP.

8. **(Weighted Bonferroni)** We wish to test m hypotheses $H_{0,1}, ..., H_{0,m}$. Let p_i be the p-value corresponding to the ith hypothesis, and let $w_1, ..., w_m \in [0,1]$ be weights such that $\sum_{i=1}^{m} w_i = 1$. Consider a *weighted Bonferroni* approach in which we reject $H_{0,i}$ when $p_i < w_i\alpha$. Show that this procedure controls FWER at level α.

9. In general, the Bonferroni method (because it uses the union bound) is conservative, but it is possible for the Bonferroni method to be *exact*. The purpose of this problem is to create a distribution of p-values for which the FWER under the Bonferroni correction is *exactly* α.

   Suppose we wish to test $H_{0,1}, ..., H_{0,m}$, all of which are truly null, and the p-values have a joint distribution that can be described as follows. Let $J \in \{1, ..., m\}$ be a random variable with $P(J = i) = \alpha/m$ for $i = 1, ..., m$, and let $p_i \in [0,1]$ depend on J:

$$p_i|(J = i) \sim Uniform(0, \alpha/m) \qquad p_i|(J \neq i) \sim Uniform(\alpha/m, 1)$$

   (a) In general, we want our p-values under the null hypothesis to have a $Uniform(0,1)$ distribution. Show that the *marginal* distribution of p_i in this setup is $p_i \sim Uniform(0,1)$.

   (b) Show that, if we reject $H_{0,i}$ when $p_i < \alpha/m$, then the FWER is exactly α.

Two-sample test for difference in means

10. Let $X_1, ..., X_n$ be an iid sample from a population with mean μ_1 and variance σ^2, and $Y_1, ..., Y_m$ an iid sample from a population with mean μ_2 and variance σ^2. The two samples are independent. Furthermore, assume that the common variance σ^2 for the two populations is finite. We wish to test the hypotheses $H_0 : \mu_1 = \mu_2$ vs. $H_A : \mu_1 \neq \mu_2$. Consider the test statistic

$$T = \frac{\overline{X} - \overline{Y}}{\sqrt{s_p^2\left(\frac{1}{n} + \frac{1}{m}\right)}}$$

    where

$$s_p^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2 + \sum_{j=1}^{m}(Y_j - \overline{Y})^2}{n + m - 2}$$

    (a) Show that if H_0 is true and $\mu_1 = \mu_2 = \mu$, you can rewrite T as

$$T = \left(\sqrt{1 - \lambda_{n,m}}\sqrt{n}(\overline{X} - \mu) - \sqrt{\lambda_{n,m}}\sqrt{m}(\overline{Y} - \mu)\right)/s_p$$

    where $\lambda_{n,m} = \dfrac{n}{n+m}$.

    (b) Suppose that $n, m \to \infty$ such that $\lambda_{n,m} \to \lambda \in (0,1)$. Using the fact that

$$\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 \xrightarrow{p} \sigma^2 \qquad \frac{1}{m}\sum_{j=1}^{m}(Y_j - \overline{Y})^2 \xrightarrow{p} \sigma^2$$

    show that $s_p^2 \xrightarrow{p} \sigma^2$.

(c) Suppose that H_0 is true and $n, m \to \infty$ such that $\lambda_{n,m} \to \lambda \in (0,1)$. Show that $T \xrightarrow{d} N(0,1)$. For this, you may use the fact that if $\{U_n\}$ and $\{V_n\}$ are two independent sequences such that $U_n \xrightarrow{d} U$ and $V_n \xrightarrow{d} V$, and U and V are also independent, then $U_n + V_n \xrightarrow{d} U + V$.

(d) Finally, give a Wald test rejection region for the test of $H_0 : \mu_1 = \mu_2$ vs. $H_A : \mu_1 \neq \mu_2$ in terms of T.

11. Consider the same setup as in the previous question, but this time also assume that the two populations are normal. That is, $X_1, ..., X_n \overset{iid}{\sim} N(\mu_1, \sigma^2)$ and $Y_1, ..., Y_m \overset{iid}{\sim} N(\mu_2, \sigma^2)$.

(a) Show that

$$\frac{(\overline{X} - \overline{Y}) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n} + \frac{1}{m}\right)}} \sim t_{n+m-2}$$

(Note: you do not need to re-derive the t distribution using Cochran's theorem here. Rather, rely on previous results from homework to show that T can be written in a way that matches the definition of a random variable following a t distribution).

(b) Give a $1 - \alpha$ confidence interval for $\mu_1 - \mu_2$ using the t_{n+m-2} distribution.

Wilcoxon signed-rank test

Let $X_1, ..., X_n$ be iid from a continuous distribution which is symmetric about θ, and let $f(x|\theta)$ denote the pdf. We are interesting in testing the hypotheses

$$H_0 : \theta = 0 \qquad H_A : \theta > 0$$

The **Wilcoxon signed-rank test** is based on the statistic

$$W_n = n^{-3/2} \sum_{i=1}^{n} R_{i,n}^{+} \text{sign}(X_i)$$

where $R_{i,n}^{+}$ is the rank of $|X_i|$ among $|X_1|, ..., |X_n|$.

The Wilcoxon signed-rank test rejects when W_n is "large". Determining a "large" value for W_n under H_0 requires finding an asymptotic distribution for W_n.

It turns out (you do not need to show this!) that

$$W_n \approx \sqrt{n}\, V_n \qquad V_n = \frac{1}{n} \sum_{i=1}^{n} U_i \text{sign}(X_i)$$

where $U_i = G(|X_i|)$ and G is the cdf of $|X|$.

11. We will find an asymptotic distribution for V_n to get an asymptotic distribution for W_n.

(a) Let $Y \geq 0$ be a non-negative continuous random variable with cdf G_Y and pdf g_Y. Show that $\mathbb{E}[G_Y^2(Y)] = 1/3$.

(b) Give an intuitive explanation that U_i and $\text{sign}(X_i)$ are independent if $H_0 : \theta = 0$ is true.

(c) Using the fact that U_i and $\text{sign}(X_i)$ are independent under H_0, show that $\mathbb{E}[U_i \text{sign}(X_i)] = 0$ and $Var(U_i \text{sign}(X_i)) = 1/3$ under H_0.

(d) Conclude that $\sqrt{n}\, V_n \xrightarrow{d} N(0, 1/3)$.

(e) Using part (d), propose a rejection region for the test of $H_0 : \theta = 0$ vs. $H_A : \theta > 0$ in terms of the statistic W_n.